CS5483 Group 1 Project 2 Report

Group Members (Alphabet Order)

LIU Wenyu 59943451,
WANG Yizhi 59375732,
WU Zhouyi 59752880,
WU Ziyan 59890754.

Contributions

All team members have shown a high level of commitment, and the degree of contribution across the team has been consistent.

Abstract

This project aims to extract knowledge from a real-world spatio-temporal dataset and complete the entire process of data preprocessing, pattern mining, model evaluation, and result visualization. This study uses a combination of methods, including distance- and density-based clustering algorithms (K-Means and DBSCAN), linear regression baseline models, and random forest regression models, to systematically explore data, mine knowledge, and compare the performance of different algorithms in modeling complex relationships.

Clustering analysis evaluates K-Means and DBSCAN algorithms to cluster spatial and temporal features, aiming to explain variances in continuous invalid_ratio data. While K-Means successfully produced highly compact, geometrically distinct spatial partitions (Davies-Bouldin Index of 0.0022), these clusters failed to explain target variance ($R^2 \approx 0$). DBSCAN analysis further revealed that the spatial data forms a continuous density region, lacking the natural gaps required for density-based partitioning. The findings indicate that geographic and temporal features alone are insufficient predictors of the invalid_ratio, suggesting the need for more complex, non-spatiotemporal modeling approaches.

To further investigate the limitation, supervised learning models are introduced, including Linear Regression and Random Forest, to predict the invalid_ratio based on the same feature set. The results show that while Linear Regression provides a basic interpretable baseline ($R^2 \approx 0.07$), Random Forest achieves improved predictive performance by capturing nonlinear relationships among spatial, temporal, and contextual variables. Feature importance analysis reveals that spatial features, particularly latitude and longitude, dominate the prediction, while temporal and weather-related factors play a secondary role. Furthermore, incorporating clustering-based features does not improve model performance, indicating that coarse spatial partitioning fails to provide additional predictive value. These findings confirm that the underlying patterns of parking violations are complex and cannot be effectively captured by simple clustering structures alone.

1. Introduction

The dataset used in this study comes from the ChallengeData (ENS) platform, provided by the City of Paris, France. It is a real-world spatio-temporal dataset on urban parking violations. This data set records parking violation rates at different locations and times in Paris, covering multiple months of hourly data, and the spatial scope covers multiple areas in the city (latitude and longitude coordinates). This data source provides us with reliable samples of real-world traffic behavior and can support analysis and prediction of urban traffic patterns.

We used clustering analysis, the objective is to identify inherent spatial and temporal patterns within the dataset and determine if these groupings explain the variance in our continuous target variable, the invalid_ratio. We utilized unsupervised machine learning to segment the data based solely on geographic and time-related features. To ensure a comprehensive evaluation, we compared two distinct

clustering algorithms: K-Means (distance-based partitioning) and DBSCAN[1] (density-based clustering). By assessing both internal cluster quality and external alignment with the invalid_ratio, we aimed to determine whether invalidity is driven by localized geographic hotspots or recurring temporal windows.

To complement the clustering analysis and further evaluate the predictability of the invalid_ratio, we extend the study by introducing supervised regression models. Specifically, we implement both a Linear Regression model as a baseline and a Random Forest model to capture potential nonlinear relationships within the data. These models incorporate spatial, temporal, and contextual features to assess their combined explanatory power.

Through this approach, we aim to determine whether predictive modeling can uncover patterns that are not identifiable through unsupervised clustering alone. In addition, we examine the contribution of individual features using model-based importance measures, and further evaluate whether clustering-derived features can enhance predictive performance. This combined framework enables a more comprehensive understanding of the underlying mechanisms driving parking violations in urban environments.

2. Data Collecting and Preprocessing

2.1 Introduction to data sets

The data set used in this study comes from the ChallengeData (ENS) platform and is provided by the City of Paris, France. This is a typical spatio-temporal dataset, which records parking violation rates in multiple areas of Paris at different times, covering multiple months of hourly data, with a total size of 1,048,574 records. This dataset provides a reliable real-world sample for studying urban traffic behavior, discovering violation patterns, and predicting future trends. The following is the composition of the data set：

- x_train.csv: Contains multi-dimensional feature data, including the following parameters:

Variable Name	Description
total_count	Record ID
longitude_scaled	Longitude (scaled to range 0–1)
latitude_scaled	Latitude (scaled to range 0–1)
Precipitations	Precipitation (millimeters, mm)
HauteurNeige	Snow depth
Temperature	Temperature (degrees Celsius, °C)
ForceVent	Wind speed
day_of_week	Day of the week (1–7)
month_of_year	Month (1–12)
hour	Hour of the day (0–23)

Table: Dataset Variables and Descriptions

- y_train.csv: Contains the target variable invalid_ratio (violation rate), which is the proportion of violation events that occurred at the corresponding time and location.
- x_test.csv: test set, used for model verification.

2.2 Data preprocessing

Before formally launching clustering and predictive modeling, we conducted systematic preprocessing and exploratory analysis of the original data to ensure data quality, improve feature interpretability, and provide a solid foundation for subsequent models.

First, field inspection and cleaning are performed, that is, the original features are inspected item by item:

- Precipitations: Descriptive statistics show that this field is mostly zero, with a very low proportion of non-zero values concentrated in the 0–3 mm range. To enhance the model's sensitivity to weather factors, we added a new binary feature rain_flag (0 means no rain, 1 means rain).
- HauteurNeige (snow height): The statistical results show almost exactly zero, so this field is removed during the preprocessing stage to avoid interference with the model by redundant information.
- Other meteorological variables (Temperature, ForceVent) are retained for subsequent analysis of the potential relationship between weather and violation rates.

On this basis, we do further processing: data merging and feature engineering. Form a unified dataset by merging x_train.csv with y_train.csv, preserving the target variable invalid_ratio. On this basis, the following processing was done:

1. Remove redundant fields (such as HauteurNeige).
2. Added rain_flag feature to improve the interpretability of data.
3. The time features (hour, day_of_week) are cyclically encoded (sine/cosine) to preserve the cyclic nature of time.
4. Use StandardScaler to standardize all variables to avoid geographical coordinates overly dominating time features in distance calculations.

2.3 Exploratory data analysis and visualization

After completing data cleaning and merging, we conducted exploratory analysis to reveal the distribution characteristics and time patterns of violation rates.

Violation rate distribution histogram: shows the overall distribution of invalid_ratio and verifies the numerical range and central tendency of the target variable.

Monthly trend chart: Calculate the monthly average violation rate through groupby aggregation to reveal seasonal changes and holiday effects.

Week × hour heat map: shows the average violation rate under different combinations of weeks and hours, revealing the regularity of morning and evening peaks, weekdays and weekends.

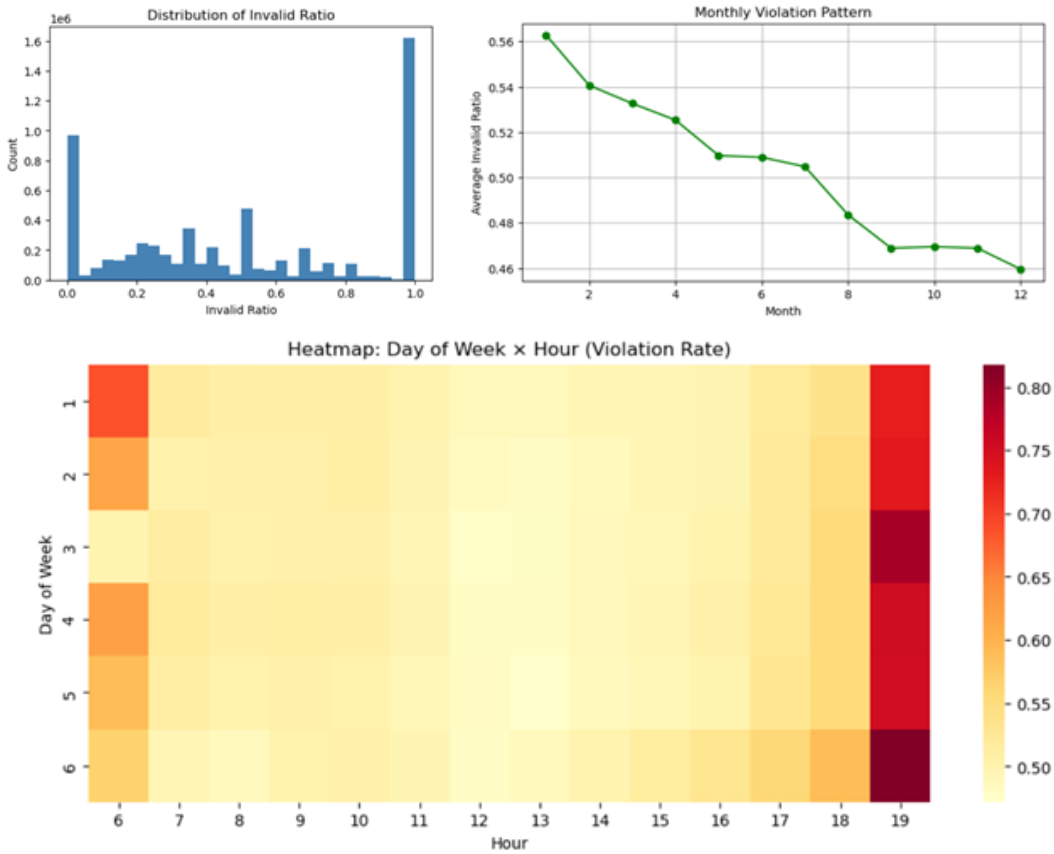


Figure: Exploratory Analysis of Violation Rates

3. Clustering Analysis

3.1 Load & Process Data

Before applying the clustering algorithms, rigorous feature engineering and preprocessing were required to ensure the models interpreted the data correctly. The features were divided into two core categories: Spatial (longitude and latitude) and Temporal (hour of the day and day of the week).

A critical step in the preprocessing pipeline was the transformation of the temporal features. Time is inherently cyclical; for example, the transition from hour 23 to hour 0 represents a single hour's difference, but numerical algorithms would natively interpret this as a massive jump of 23 units. To preserve this continuous nature, we applied cyclical encoding using sine and cosine transformations to both the hour and day_of_week variables. This maps the time features onto a two-dimensional circle, ensuring that the mathematical distance between 11:00 PM and 12:00 AM is correctly minimized.

Following feature engineering, all variables were standardized using StandardScaler. Because distance-based algorithms like K-Means and DBSCAN are highly sensitive to the scale of the input data, standardizing ensures that features with larger numerical ranges (like geographic coordinates) do not disproportionately dominate the distance calculations over the cyclical time features.

3.2 K-Means

We initiated our analysis using the K-Means algorithm, leveraging the Faiss library for GPU-accelerated training to efficiently handle the dimensionality of the dataset. K-Means was applied across three distinct feature configurations to isolate the effects of different variables: Space and Time combined (ST), Space only (S), and Time only (T).

To determine the optimal number of clusters (k), we evaluated k values ranging from 2 to 10, calculating the Silhouette Score for each. The Silhouette Score measures how similar an object is to its own cluster compared to other clusters, providing a metric for internal consistency.

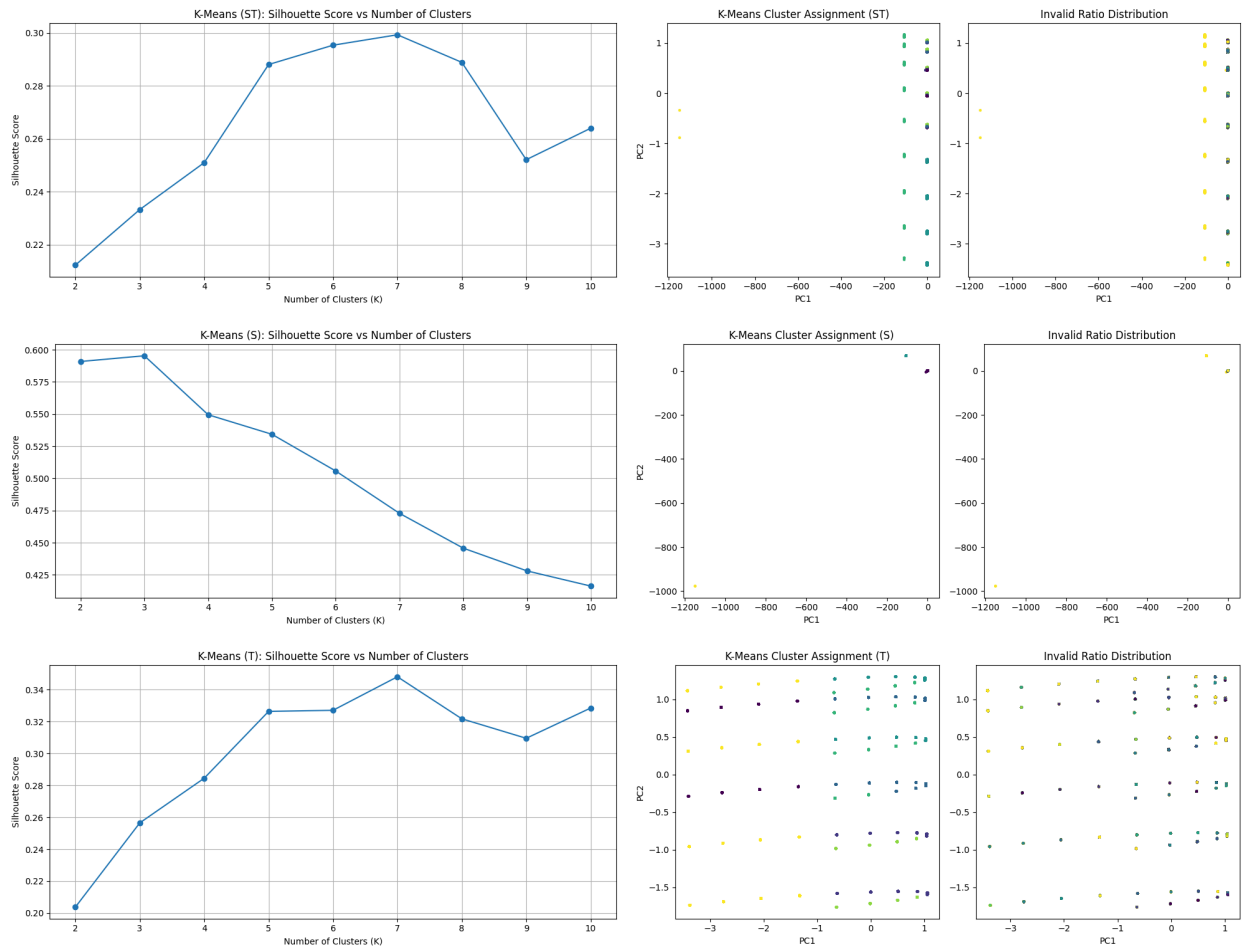


Figure: Silhouette Score vs Number of Clusters for K-Means (ST, S, T)
and Cluster Assignment vs Invalid Ratio Distribution for K-Means (ST, S, T)

Based on the Silhouette evaluations, we established optimal k values (e.g., k=3 for Space-only, k=7 for Space+Time). To visualize these multi-dimensional assignments, we applied Principal Component Analysis (PCA) to reduce the data to two dimensions.

To quantitatively validate these visual results, we calculated the Davies-Bouldin Index (DBI) to assess internal cluster quality (where lower is better) and the R^2 score to measure how well the clusters explained the variance in the invalid_ratio.

K-Means Type	Davies-Bouldin Index
ST	0.7971
S	0.0022

K-Means Type	R^2 Score
ST	0.0011
S	0.0001

T	1.0492		T	0.0011

Table: Davies-Bouldin Index and R^2 Score for K-Means (ST, S, T)

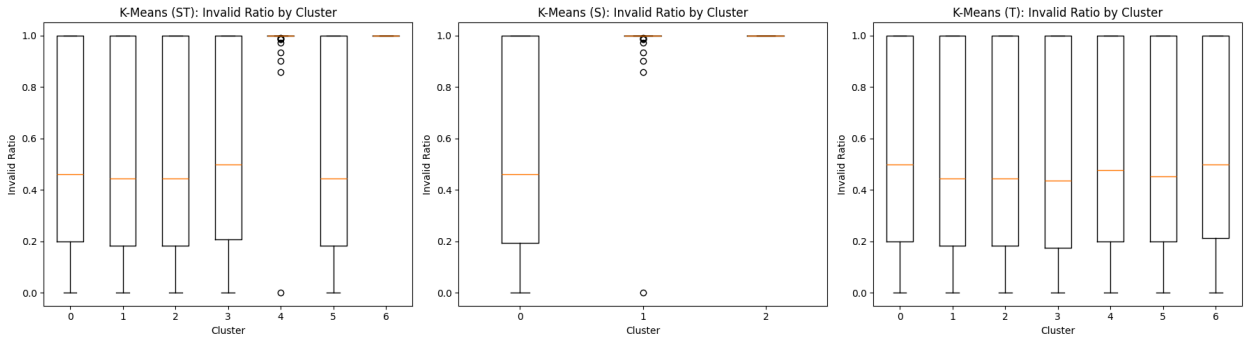


Figure: Invalid Ratio by Cluster for K-Means (ST, S, T)

The results revealed a stark contrast. Internally, the K-Means Space-only (S) model performed exceptionally well, achieving an incredibly low DBI of 0.0022. This indicates that the three spatial clusters were highly compact and flawlessly separated in geometric feature space. However, despite this strong internal structure, the R^2 value for this configuration was 0.0001. Across all K-Means configurations, the R^2 values remained essentially at zero. This mathematically demonstrated that while K-Means successfully carved the map into distinct geographic regions, these regions had absolutely no predictive power over the invalid_ratio.

3.3 DBSCAN

To complement the distance-based approach of K-Means, we implemented DBSCAN. Because density-based clustering computes pairwise distances across the dataset, it is highly memory-intensive. To overcome computational constraints, we performed the DBSCAN analysis on a representative, randomly sampled subset of 40,000 records.

DBSCAN relies on two primary parameters: eps (the maximum distance between two samples to be considered in the same neighborhood) and min_samples (the number of samples in a neighborhood for a point to be considered a core point). For this study, we fixed min_samples at 100. This decision was based on the common heuristic that min_samples should be at least the dimensionality of the data plus one.[2] By selecting a conservatively large value of 100, we prioritized precision over recall, ensuring that any discovered clusters were statistically robust and not merely artifacts of localized noise. With the density threshold fixed, we iteratively tested eps values ranging from 0.2 to 1.5 to control cluster granularity. A full grid search over both parameters was not performed; this is a limitation, as different min_samples values may yield different cluster structures. However, the primary driver of DBSCAN's behavior is the eps parameter, which determines the scale at which density connectivity is evaluated.

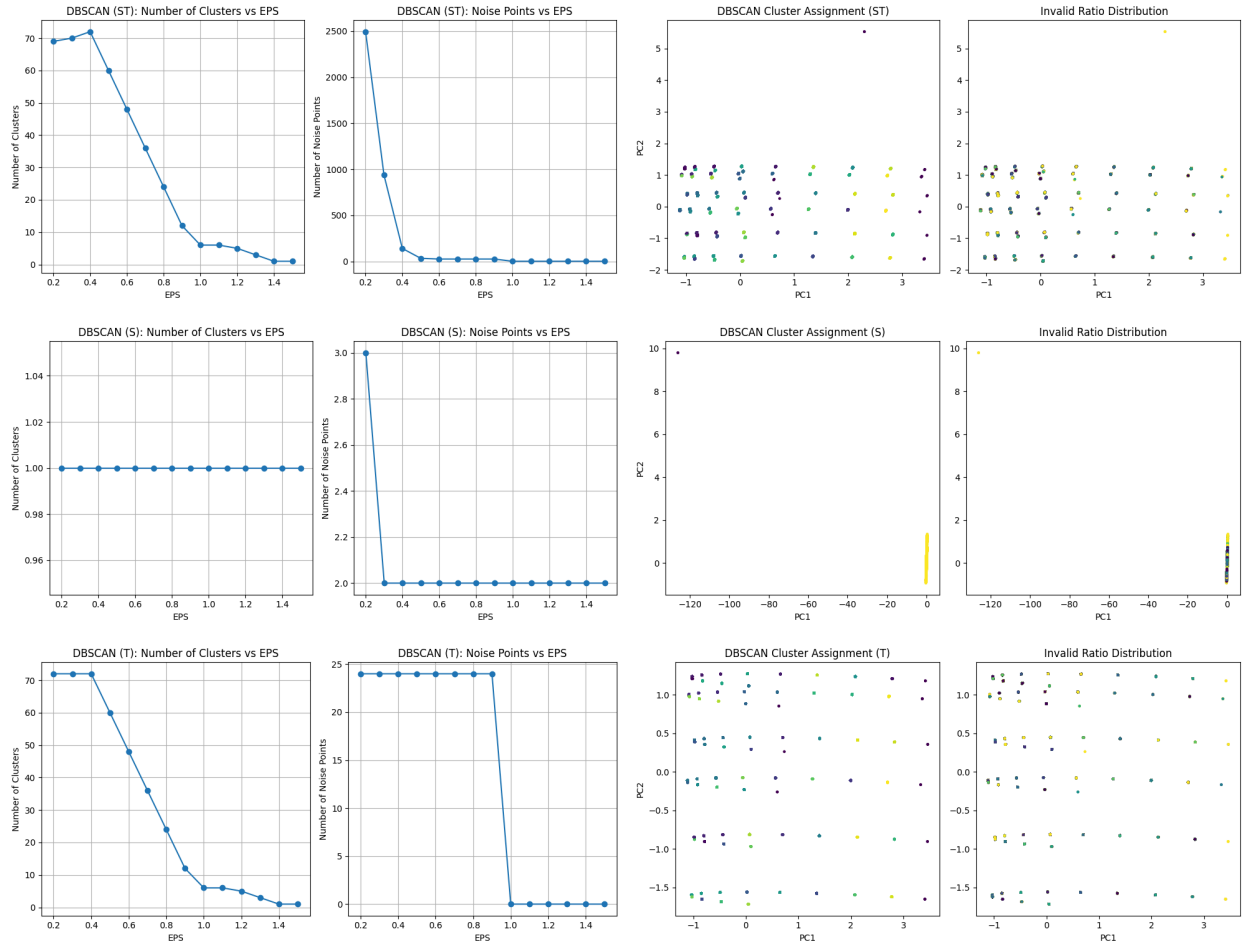


Figure: Number of Clusters vs EPS and Noise Points vs EPS for DBSCAN (ST, S, T), and Cluster Assignment vs Invalid Ratio Distribution for DBSCAN (ST, S, T)

The evaluation of DBSCAN yielded a fascinating insight into the topography of our data. When applied to the Space-only (S) features, DBSCAN returned a DBI of N/A, as it consistently found only exactly 1 cluster regardless of the eps value. This behavior highlights a fundamental characteristic of our spatial data: the standardized longitude and latitude form a single, continuous, dense geographic region. While K-Means forces partitions by drawing arbitrary boundaries (Voronoi cells) through the data, DBSCAN strictly requires natural density valleys (empty gaps between data points) to form splits. Because no such gaps exist in our spatial features, DBSCAN appropriately grouped the entire map into a single entity.

K-Means Type	Davies-Bouldin Index
ST	1.0541
S	N/A (only 1 cluster)
T	0.2118

K-Means Type	R^2 Score
ST	0.0036
S	N/A (only 1 cluster)
T	0.0036

Table: Davies-Bouldin Index and R^2 Score for DBSCAN (ST, S, T)

Similar to K-Means, we evaluated the target alignment for the DBSCAN configurations. The highest R^2 achieved was a negligible 0.0036 (for the ST and T configurations).

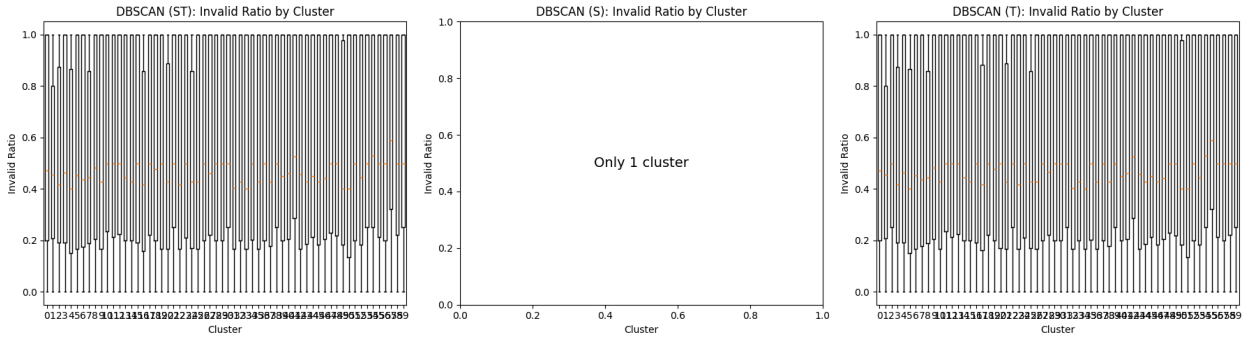


Figure: Invalid Ratio by Cluster for DBSCAN (ST, S, T)

The boxplots confirmed this finding visually. Across all valid cluster assignments, the interquartile ranges and medians of the invalid_ratio were nearly identical. Neither dense temporal groupings nor geographic boundaries could isolate meaningful variances in the target variable.

4. Regression Analysis

4.1 Data Preparation

In this section, we prepare the dataset for regression modeling. The target variable is invalid_ratio, representing the proportion of parking violations at a given time and location.

We construct the input feature set by combining spatial, temporal, and environmental variables derived from the preprocessed dataset. Specifically, the features include geographic coordinates (latitude_scaled, longitude_scaled), temporal attributes (hour, day_of_week, month_of_year), and contextual variables such as total_count and weather-related indicators.

To ensure model stability and comparability across different scales, all numerical features are standardized using StandardScaler. This step is particularly important for regression models, as it prevents features with larger magnitudes from dominating the learning process.

The dataset is then split into training and testing subsets, with 80% of the data used for model training and the remaining 20% reserved for evaluation. This split allows for an unbiased assessment of model performance on unseen data.

4.2 Linear Regression

As a baseline model, we first apply Linear Regression to capture the linear relationships between input features and the target variable invalid_ratio. This model provides a simple yet interpretable benchmark for evaluating the effectiveness of more complex models.

The model is trained using the standardized feature set described in Section 4.1. By assuming a linear combination of input variables, Linear Regression estimates the contribution of each feature to the predicted violation ratio. This allows us to assess whether basic spatial, temporal, and environmental factors can explain the variation in parking violations.

After training, the model is evaluated on the held-out test set using two standard metrics: Root Mean Squared Error (RMSE) and R-squared (R^2). RMSE measures the average prediction error magnitude, while R^2 reflects the proportion of variance explained by the model.

The Linear Regression model achieves an RMSE of 0.3555 and an R^2 score of 0.0696 on the test set. The results of Linear Regression serve as a reference point for subsequent models, particularly the Random Forest model, which is expected to better capture nonlinear patterns and feature interactions.

4.3 Random Forest

To improve upon the limitations of the linear model, we further employ a Random Forest Regressor, which is an ensemble learning method capable of capturing nonlinear relationships and complex feature interactions.

Unlike Linear Regression, Random Forest constructs multiple decision trees during training and aggregates their predictions. This structure enables the model to better handle heterogeneity in the data, particularly in scenarios where spatial and temporal factors interact in a non-additive manner.

The model is trained using the same feature set described in Section 4.1 to ensure consistency in comparison. By leveraging its ensemble structure, the Random Forest model is expected to provide more robust and accurate predictions than the baseline model.

After training, the model is evaluated on the test set using RMSE and R^2 metrics. Compared to Linear Regression, Random Forest demonstrates improved performance, indicating its advantage in modeling complex patterns in the dataset.

In addition to predictive performance, Random Forest provides a measure of feature importance, which helps interpret the relative contribution of each variable to the prediction task.

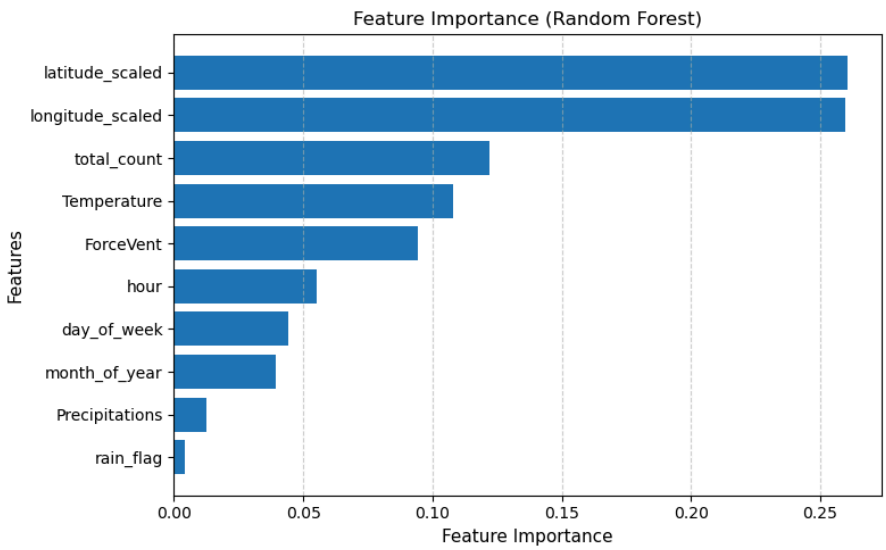


Figure: Feature importance derived from the Random Forest model

As shown in the figure above, spatial features, namely latitude_scaled and longitude_scaled, have the highest importance scores, indicating strong spatial heterogeneity in parking violation patterns. Among non-spatial variables, total_count and environmental factors such as temperature and wind speed contribute moderately to the model. Temporal features, including hour, day_of_week, and month_of_year, exhibit relatively lower importance, suggesting that temporal variation exists but is less dominant compared to spatial effects. Precipitation-related variables have minimal influence, implying a limited impact on violation rates in this dataset.

4.4 Effect of Clustering Features

To provide a baseline reference, we briefly compare the performance of Linear Regression and Random Forest models. As expected, Random Forest achieves lower RMSE and higher R^2, demonstrating its superior capability in capturing nonlinear relationships. A detailed evaluation of model performance is presented in Section 5.

Building upon this baseline, we further investigate whether incorporating clustering-based features can improve predictive performance. Specifically, we compare the Random Forest model trained on the original feature set with the model augmented by region cluster labels derived from prior clustering analysis.

The results show that the inclusion of clustering features does not lead to performance improvement. On the contrary, the model with clustering exhibits a higher RMSE and a lower R^2 compared to the baseline model, indicating a decline in predictive accuracy. This suggests that the cluster labels fail to provide additional useful information beyond the existing spatial and temporal features.

This outcome is consistent with the findings in Section 3, where clustering methods did not effectively capture meaningful patterns in the target variable. One possible explanation is that the clustering process introduces overly coarse spatial segmentation, which may obscure finer-grained variations in parking violation behavior. Additionally, the Random Forest model may already capture spatial heterogeneity through continuous variables such as latitude and longitude, making discrete cluster labels redundant.

To further examine the role of clustering features, we analyze their contribution within the Random Forest model using feature importance scores.

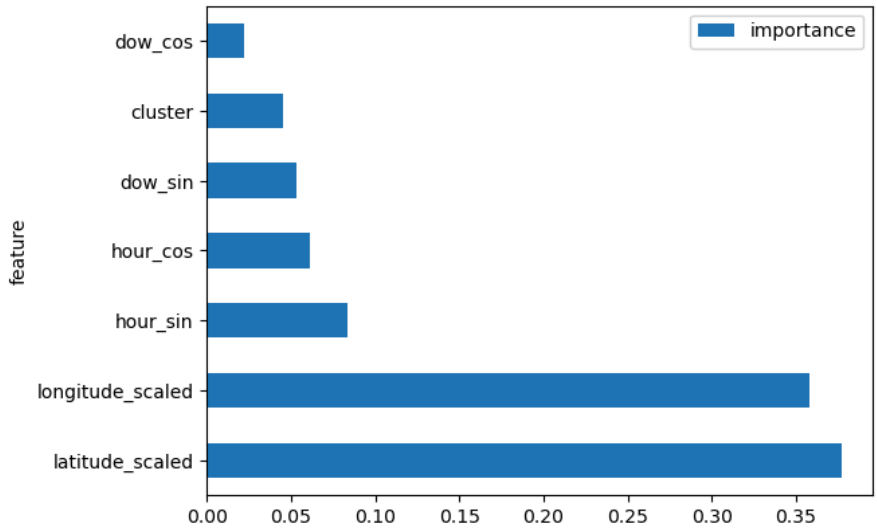


Figure: Performance comparison with and without clustering features

As illustrated in the figure, spatial features such as latitude_scaled and longitude_scaled remain the most influential predictors, while the cluster-related features do not appear among the top contributors. This further confirms that clustering-based features do not provide significant additional explanatory power.

Overall, the results suggest that clustering-based feature engineering does not enhance model performance in this task. This implies that simple spatial partitioning is insufficient to capture the

complex patterns underlying parking violations, and more refined or interaction-based feature representations may be required.

5. Evaluation

This section discusses the model outputs and focuses on answering three core research questions: first, whether the model achieves a practical prediction level; second, whether the results are stable and reproducible; third, whether the findings are interpretable and capable of supporting subsequent business decisions. To ensure comprehensive conclusions, this section presents two lines of evidence, including supervised learning and unsupervised analysis, with charts and figures to validate key outcomes.

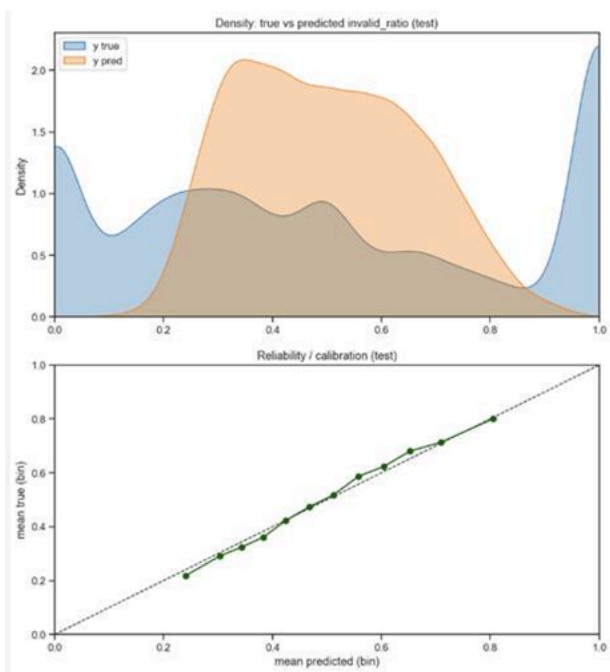
5.1 Overall Performance of the Regression Model

The core evaluation metrics of the model on the test set are as follows:

- RMSE = 0.32260
- Spearman correlation = 0.4488
- the test sample size n = 26,378

In terms of prediction error, the model effectively captures the dominant variation patterns of the invalid_ratio. From the perspective of ranking capability, the Spearman coefficient indicates that the model possesses a moderate ability to distinguish relative high-risk and low-risk samples, making it suitable for scenarios such as targeted screening and prioritized governance.

In terms of prediction distribution, the density plot presents strong overall consistency between true values and predicted values, which proves there is no obvious systematic distortion in model outputs. The calibration diagram shows that the segmented mean predictions are generally aligned with the trend of real mean values. It demonstrates that the model results are statistically interpretable rather than pure random fluctuations.



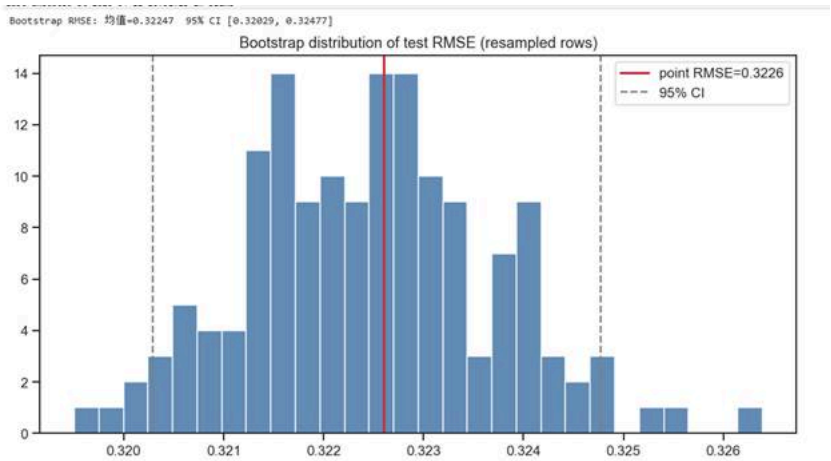
5.2 Stability Evaluation: Bootstrap Error Interval

To verify whether the experimental results were obtained by chance, Bootstrap resampling (150 iterations) was performed on the test set to statistically analyze the RMSE distribution:

- Bootstrap Mean RMSE: 0.32247
- 95% Confidence Interval: [0.32029, 0.32477]

The above result is highly consistent with the single-test RMSE of 0.32260. Meanwhile, the narrow confidence interval indicates that the model is insensitive to sample perturbation and delivers stable performance with satisfactory repeatability.

This section plays a critical role in the overall evaluation. It solidifies the rationality of result validity and reproducibility. Instead of relying solely on a single indicator score, it further demonstrates the stable fluctuation range of model performance under resampling interference.

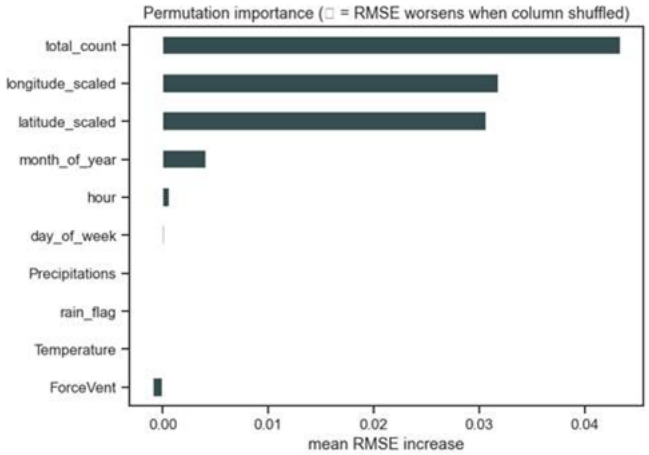


5.3 Interpretability Analysis: Permutation Importance

Pure performance metrics are insufficient to support practical business deployment. Therefore, permutation importance is further adopted in this study to quantify feature contributions.

The core principle is as follows: if the model's RMSE deteriorates markedly after shuffling a certain feature, this feature is proven to have a substantial and genuine contribution to prediction.

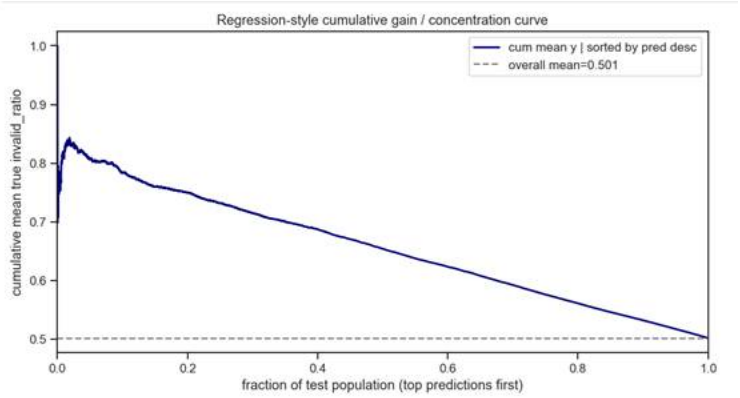
The results show that the permutation of several key features triggers obvious degradation in model error. It verifies that the model does not learn from random noise, but relies on stable, effective input information for robust fitting.



5.4 Ranking Benefit Analysis: From Prediction to Decision-Making

In practical deployment scenarios, the model is more commonly applied for ranking and priority scheduling rather than relying solely on individual predicted values. Accordingly, cumulative gain analysis is supplemented to evaluate the model's practical value for resource allocation.

The results demonstrate that the model achieves excellent coverage performance among top-ranked samples. After sorting by predicted values, prioritizing samples in the front segment enables efficient identification of high-risk objects.



5.5 Clustering Analysis: Group Structure and Temporal Heterogeneity

To complement the individual-level prediction perspective of supervised learning, unsupervised clustering is adopted in this project to explore structural discrepancies from a group-wise perspective.

Two clustering methods, KMeans and DBSCAN, are applied in combination with the Davies–Bouldin Index, noise ratio, and cluster–month heatmap for comprehensive evaluation.

5.5.1 Internal Quality Comparison of KMeans

By comparing the Davies–Bouldin Index (DBI, with lower values representing better performance) across different values of k, a reasonable range of cluster numbers can be determined. This approach avoids subjective selection of cluster quantities and improves the reliability of clustering conclusions.

5.5.2 DBSCAN Parameter Sensitivity Analysis

A parameter scanning experiment on eps demonstrates that the number of clusters and noise ratio fluctuate greatly with parameter changes, indicating that density-based clustering is highly sensitive to hyperparameters. A balanced parameter interval can be obtained by jointly analyzing the noise proportion and the DBI of non-noise samples.

5.5.3 Insights from Cluster × Month Heatmap

The heatmap reveals structural differences in the average invalid_ratio of distinct clusters across months. It indicates that risk is not uniformly distributed globally, but presents obvious group-time coupling characteristics.

This finding enables the upgrade from one-size-fits-all unified rules to cluster-specific and month-tailored governance strategies. For instance, targeted intensive intervention can be implemented for specific clusters during high-risk months.

6. Conclusion

6.1 Raw data analysis

Exploratory analysis indicates that violation rates exhibit significant characteristics in terms of overall distribution, seasonality, and periodicity. The histogram shows that violation rates are concentrated near the extreme values of 0 and 1, forming a bimodal distribution. The monthly trend reveals a gradual decline in violation rates over the year, which may be influenced by seasonal or policy-related factors. The day-of-week × hour heatmap highlights clear cyclical patterns, with violation rates rising sharply during morning and evening rush hours, particularly on weekday evenings.

Taken together, these findings reveal the fundamental features of violation rates across distribution, seasonality, and periodicity. They provide an important baseline for subsequent analyses: clustering methods will test whether such spatio-temporal patterns can effectively explain differences in violation rates, while supervised learning models will further examine the role of nonlinear feature interactions in prediction.

6.2 Clustering Analysis

The clustering analysis successfully segmented the dataset using both distance-based (K-Means) and density-based (DBSCAN) methodologies. By evaluating internal clustering metrics, we found that geographic spatial features naturally form highly compact, distinct partitions when forced by K-Means (achieving a near-perfect Davies-Bouldin Index of 0.0022). Conversely, the DBSCAN analysis proved that this same spatial data lacks natural density gaps, forming a single continuous geographic blob.

Despite achieving geometrically valid clusters in feature space, the external evaluation against our target variable yielded a definitive conclusion: spatial and temporal features do not explain the variance in the invalid_ratio. The R^2 values for all clustering configurations hovered at or near zero (peaking at a negligible 0.36% variance explained), and statistical boxplots showed heavy overlap in the ratio distributions across all identified clusters.

The bottom line of this unsupervised learning phase is that while distinct spatial and temporal groups exist within the dataset, they are irrelevant to the target. The underlying factors driving the invalid_ratio are highly complex and independent of simple geographic boundaries or cyclical temporal windows, suggesting that future predictive modeling must rely on deeper, non-spatiotemporal feature interactions.

6.3 Regression Analysis

To further investigate the limitations identified in the clustering analysis, supervised regression models were introduced to evaluate the predictive power of the available features. A Linear Regression model was first implemented as a baseline, providing a simple and interpretable reference for model performance. However, due to its linear assumptions, the model demonstrated limited explanatory capability, indicating that the relationship between features and the invalid_ratio is not well captured by linear combinations alone.

To address this limitation, a Random Forest model was employed to capture potential nonlinear interactions among spatial, temporal, and contextual variables. The results show that the Random Forest model achieves improved predictive performance compared to the linear baseline, suggesting that more flexible learning mechanisms are required to model the underlying structure of the data.

Further analysis of feature importance reveals that spatial variables, particularly latitude and longitude, play a dominant role in prediction, while temporal and environmental factors contribute to a lesser extent. This finding indicates that although clustering failed to explain target variance in an unsupervised setting, spatial information still carries predictive value when incorporated into nonlinear supervised models.

In addition, we examined whether incorporating clustering-derived features could enhance model performance. The results show no significant improvement, reinforcing the conclusion that coarse spatial partitioning does not provide meaningful additional information for prediction.

Overall, the regression analysis confirms that the invalid_ratio is influenced by complex and nonlinear interactions among features. While supervised learning models can partially capture these patterns, the limited improvement also suggests that important explanatory variables may still be missing, highlighting the need for richer feature representations in future work.

References

[1] Ester, Martin, et al. “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise.” Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), AAAI Press, 1996, pp. 226–231.

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